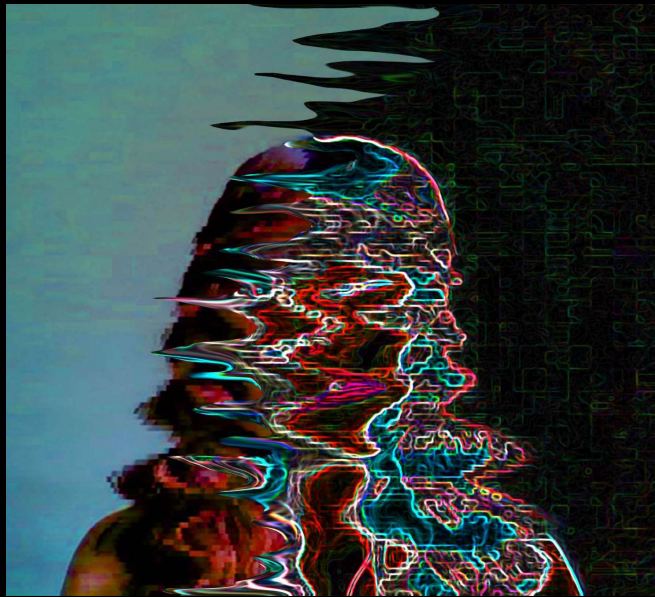




YEN-YI, HUANG

PORTFOLIO

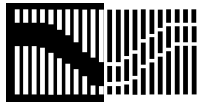
Integrated Digital Media
New York University

CONTENT



01 Metamorphosis

Interactive fabrication: from body motion to three-dimensional models P.4 - 9



02 Seizing the Ever-Changing Light

Re-Experiencing the Visual World by Programming the City P.10 - 14



03 Melting into River

Interactive Design Expressing the Emotion of Separationnn P.15 - 18



04 Bubble Up

Dome Flagship of Gas Station P.19 - 22



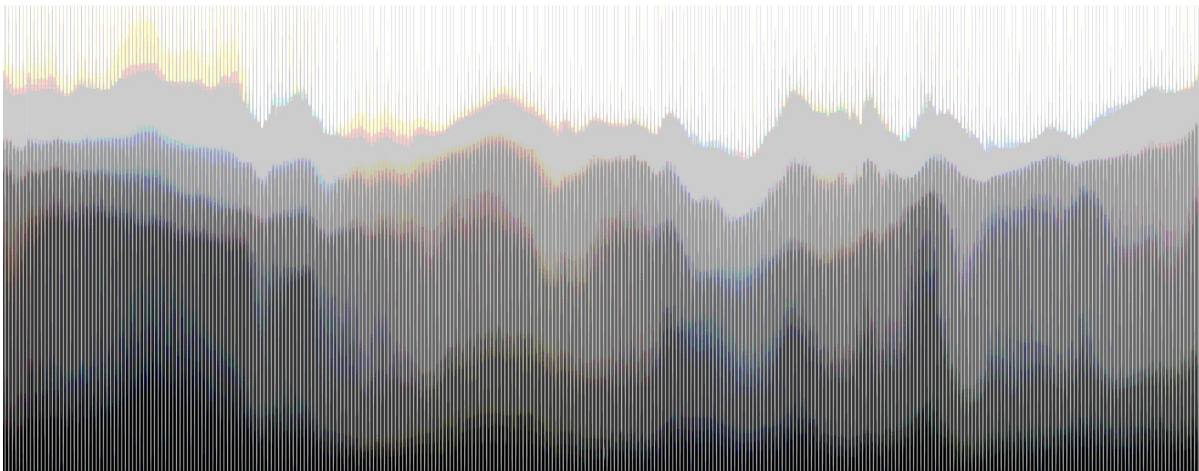
05 1 * 3

2016 Student Summer Seminar Structural Design Competition P.23



06 Explorations of Plans Generator by Machine Learning

..... P.24



Intro

For me, architecture is not only about creating a tangible building but also designing the relationship between humans and the external world.

I believe that the world contains diverse information, both visible and invisible. I aspire to explore more about the world using innovative technology and create new designs in order to help people better understand and have a deeper connection with the world around them.

01

Metamorphosis

Interactive fabrication: from body motion to three-dimensional models

Yen-ting Cho Studio

Programmer | Yen-ting, Yeh Yen-Ling, Kuo

Computational Designer | Yen-Yi, Huang

Advisor | Yen-Ting Cho, choyenting@mail.ncku.edu.tw

Technique | Simulation on Rhino & Grasshopper, Motion Capture by Kinect

URL | yenyihuang.com/work/metamorphosis

The project aimed to make participants project themselves onto the alien form. By observing the rules and feedback of the form, the participants may change the way they used to move their bodies. In general, there is a convention that how a normal human acts. For example, people move on their legs, not on hands, knees, or any parts of the body. During the interaction process, the participants could concentrate on their body movement and be inspired to have different knowing and imagination to their body.

Dancer: Shih-Ting Yen



The Transformation of Different Body Skeletons

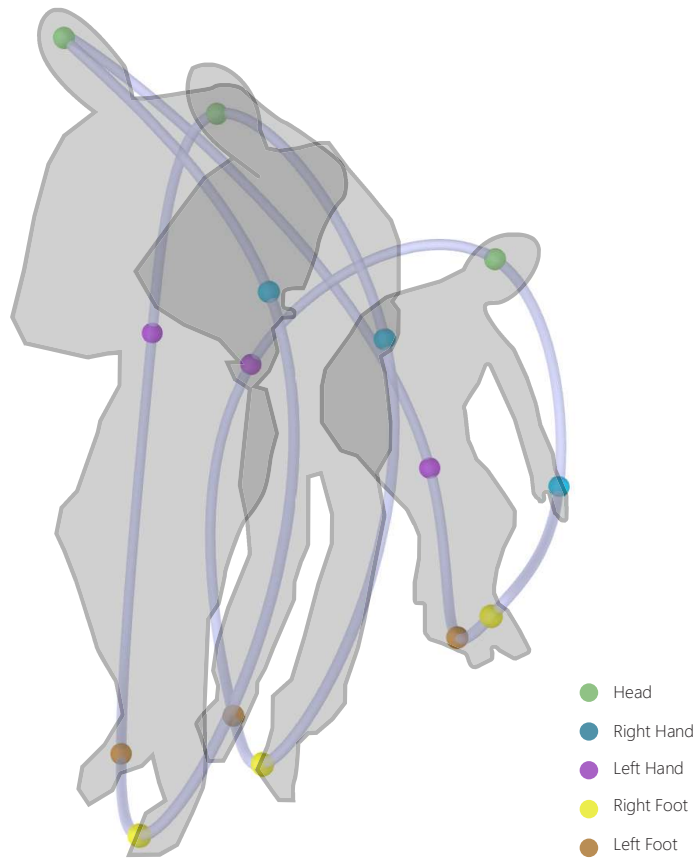


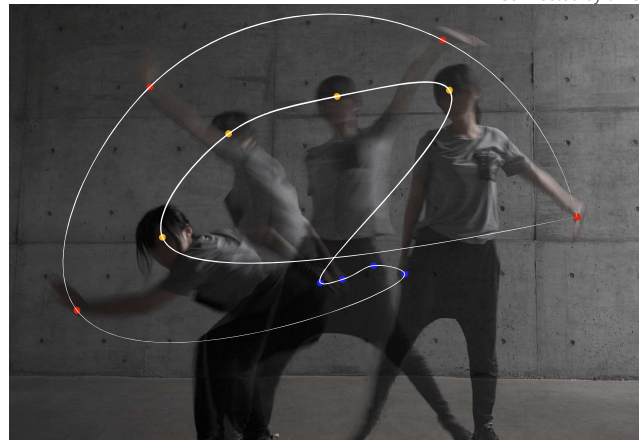
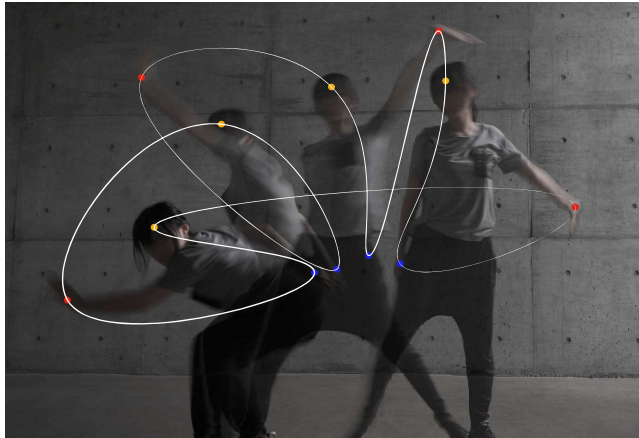
Fig. The relationship of the transformation model and body skeletons

	5 Points : Whole Body	1 Point : Left Hand	2 Points : Hands	3 Points : Head & Feet
Body Skeletons				

1. Form Development

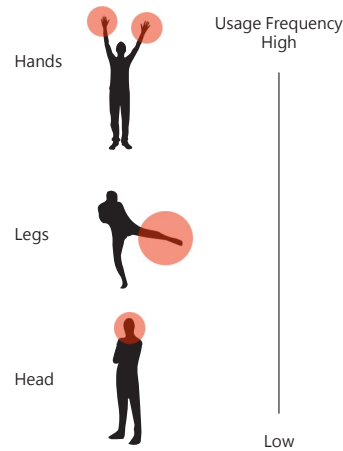
From input data to model

Kinect captured 17 points from the human skeletons. Due to the processing speed, we used the most efficient points: head, palms and feet. In the physical world, the four limbs all connect to the torso. In the virtual world, the extracted five points deconstruct the original organization of the human body. I developed several geometrical forms to reconstruct the relationship between the points. Although the relationship of the 5 points is reorganized, the forms still are influenced by the origin relationship of the human body from explicit to implicit.



Connected by parts
Fig. Two Rules Connecting Body Skeletons

3 Categories of Usage Frequency



Same Data Overlap

Skeletons						
Overlap						
connected by parts						
connected by time						

Different Data Overlap

Skeletons								
Different Data Same Rule	connected by parts							
	connected by time							
Different Data Different Rule	Red: by parts Blue: by time							
	Red: by time Blue: by parts							

The preference for action

I found that people's frequency of using specific body parts is quite different. People tend to use their hands more than their legs, the last is their head. I classify the five body points into 3 categories by the preference of action. Based on the logic, I tried to combine the tubes in diverse possibilities from different categories and got six groups of points. Then I chose two of the six groups to combine and remove the ones overlapping over two points.

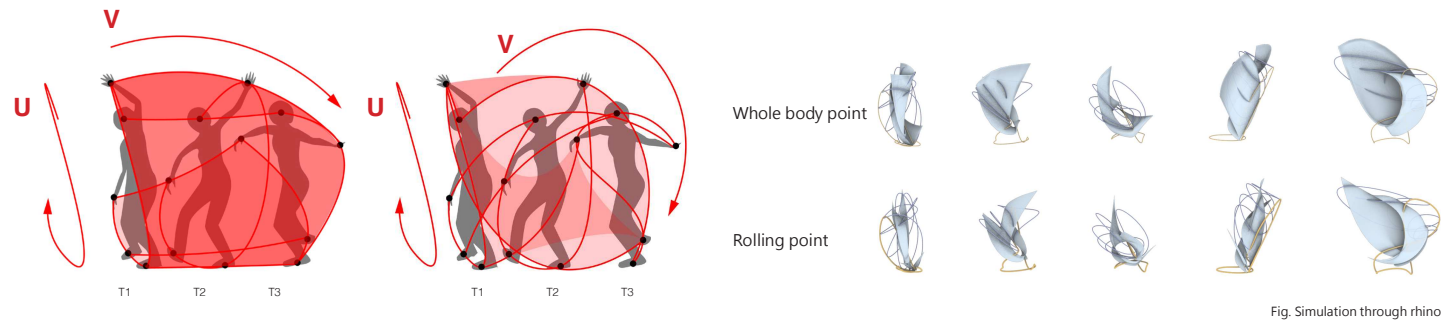
1-1. line

To connect the points, I developed two rules for drawing a curve: by time and by parts. The form of the rule drawing curve by time would be several rounds which emphasizes the frames of time. It is instinctive for the participant to sense the change and the rule through the interaction. The rule of drawing curves by parts would connect the points in the sequence of different human parts over time. The form would be much simpler than the form of time rule and implicit to sense by the participants.

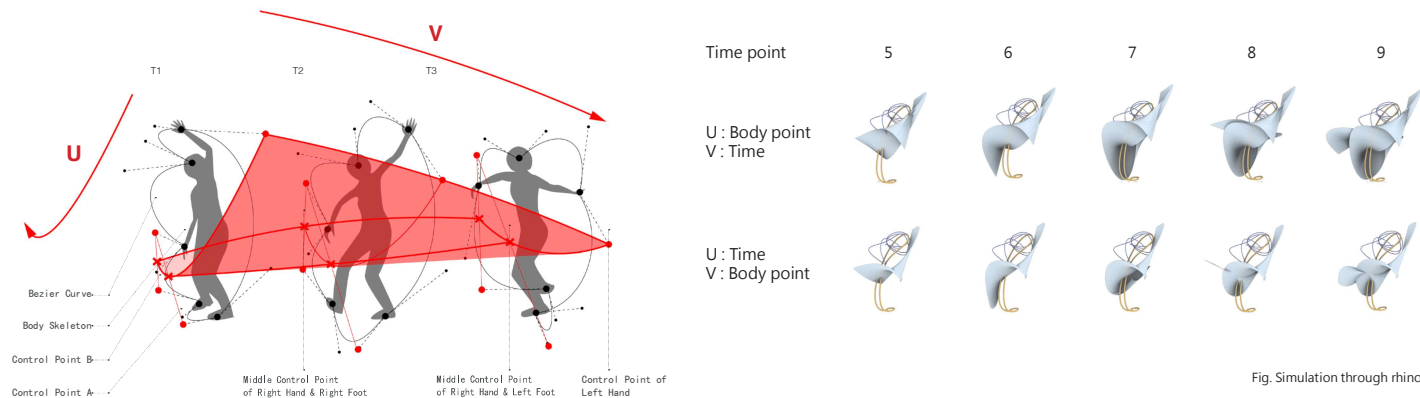
Points and Lines to the Surface

Skeletons	2 Points : Hands			3 Points : Head & Hand & Foot					
	Straight Line			2 Straight Lines		Curve : Open End		Curve : Close End	
Control Line									
Loft Surface									
Fit Loft Surface									

Points to the Surface: Whole body points to rolling body points



Points to the Surface: Extended Surface



1-2. Surface

Points and Lines to the Surface

I used the points to create the lines, then used the lines to loft a surface. There are two ways to create a line : 2 points to a straight line, 3 points to 2 straight lines or a curve. It is interesting that the surfaces made by straight lines has sharp shape differ from the curves.

Points to the Surface

I used whole body points in the sequence from the head, left hand, left foot, right foot and right hand to organize the U vector, and use the time to be the V vector. The form would be a U-shaped surface. To create a complicated surface, I shift the input data by time. In the original sequence, the head would always be the first data. After shifting the data, the first data in different moments will roll. It created a twist and sharp-edged surface.



Fig. Surface : whole body points in live interaction



Fig. Surface : rolling body points in live interaction



Fig. Surface : Extended in live interaction

2. Fabrication

The matrix shows the development of the "interactive fabrications", which somehow looks like an evolution of a species to me.

WHOLE TIME SPAN AUDITION

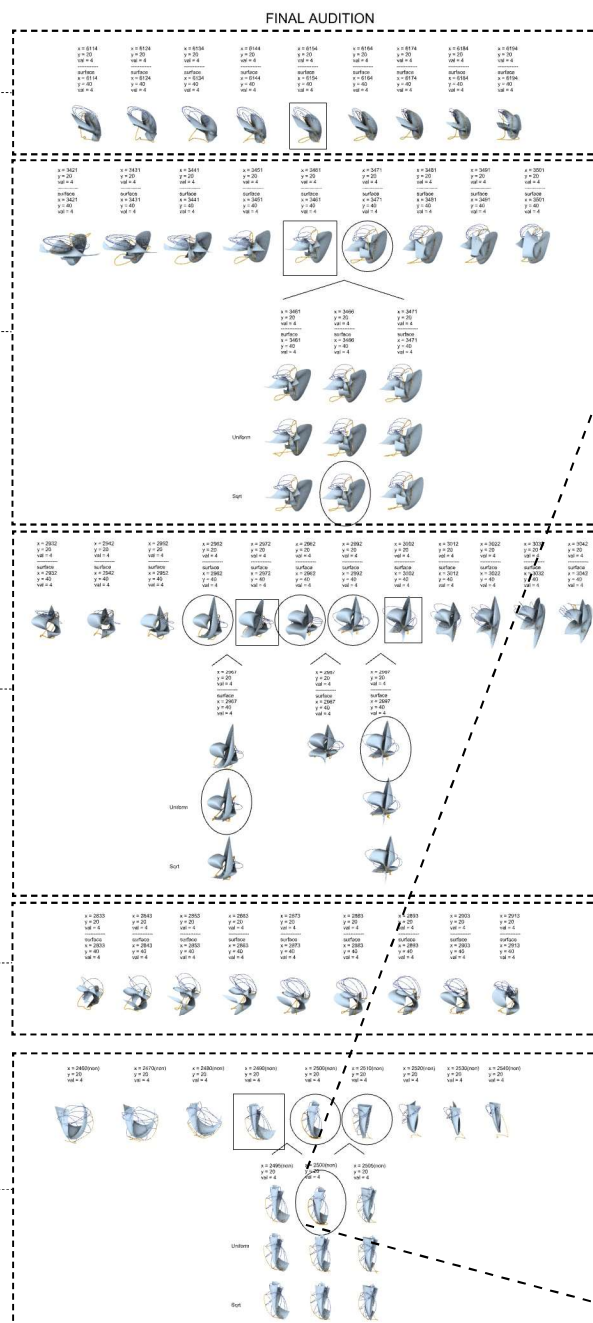
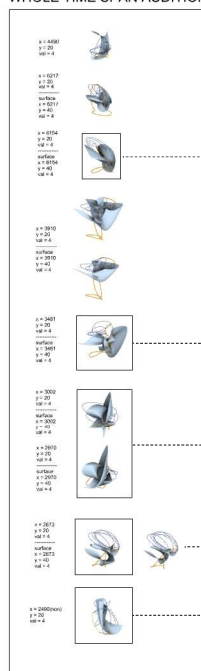


Fig. Choosing process to fabrication

Fig. 3D printed model 8

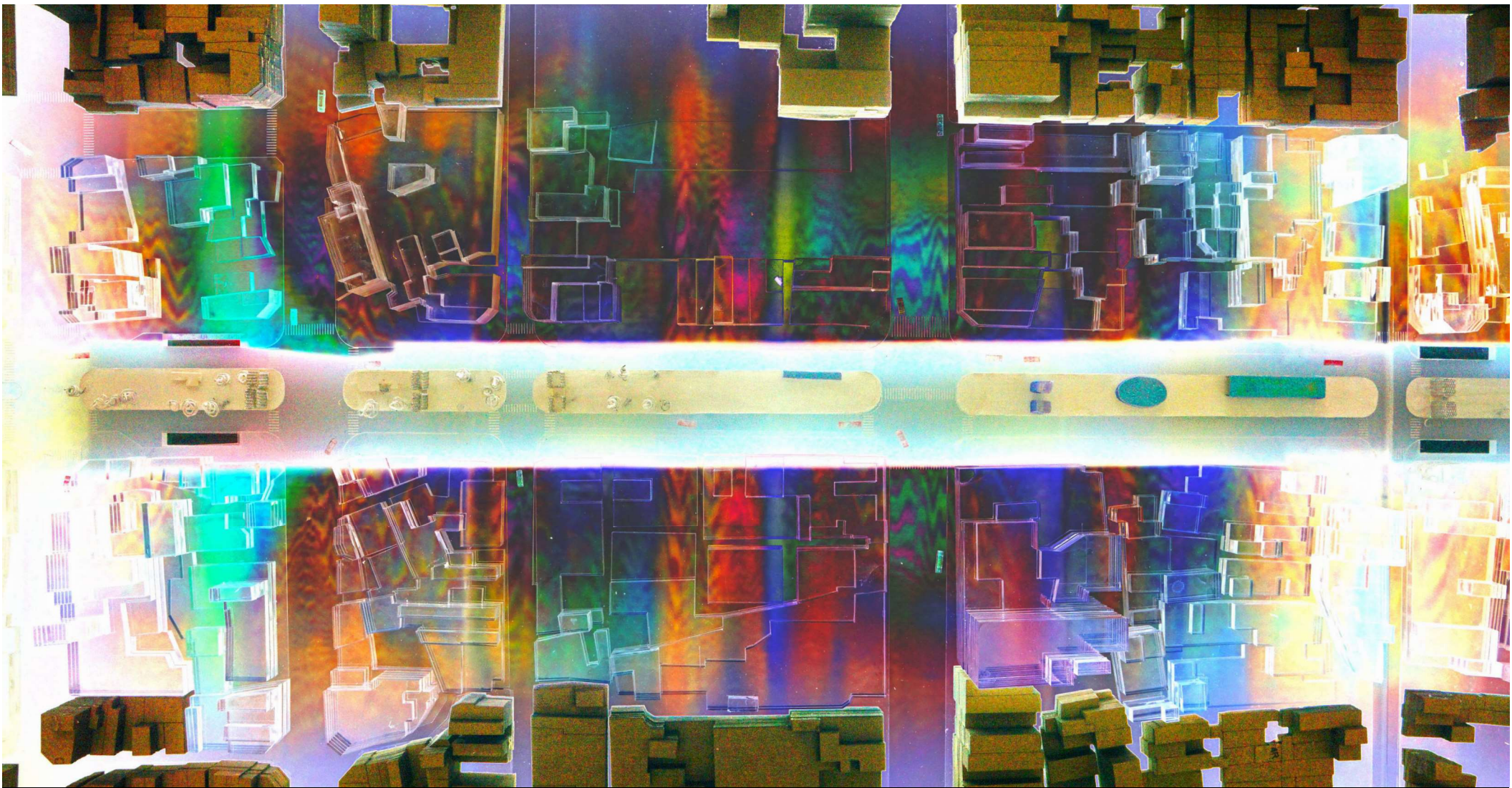
3. Live Interaction

The virtual model shown on the monitor will interact with the dancer's motion simultaneously, and also influence the dancer's next move. The interactive process, therefore, forms a dynamic dialogue between the virtual and real world, the intangible and the tangible.

Video: yenyihuang.com/work/metamorphosis



Fig. The live Interactive process Fig. Dancer interacted with the model 9



02

Seizing the Ever-Changing Light

Re-Experiencing the Visual World
by Programming the City

Site | Haian Road, Tainan
Type | Thesis Project, Individual Project
Advisor | Cheng-Luen Hsueh, clhsueh@mail.ncku.edu.tw
Technique | Java on Processing

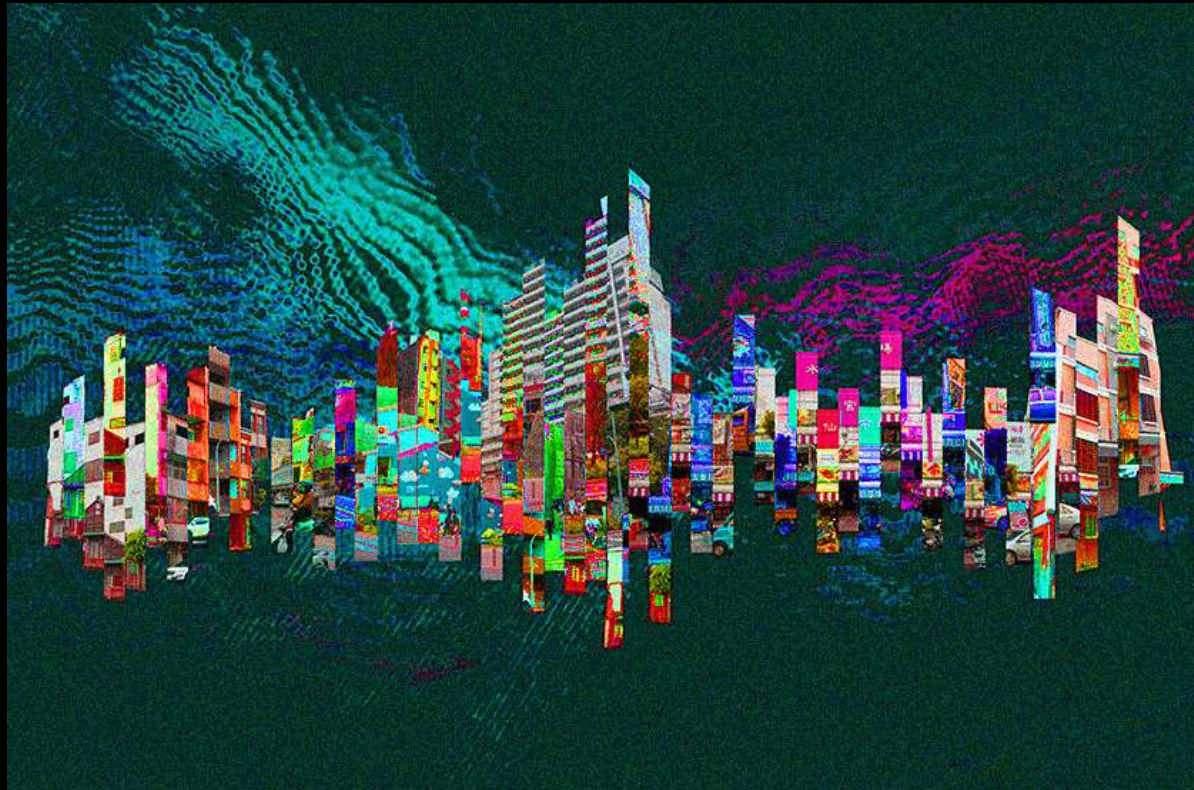


Fig. Concept diagram
Fragmentation of one's urban perception

1. Intension

The cognition of the city

According to Kevin Lynch, two factors are building up the legibility of the city: immediate sensation and the memory of the experience. To influence people's cognition of the city, I tried to add another layer on the sensation of the environment by changing the vision of the city.

Extraction of the information

To break the cognition of a symbol, I tried to blur the contour of the object. People could still tell the scene and the viewer's perspective by color and light. Then, After removing the color, people could only sense the scene simply by the brightness of the frame and figure it out by projecting similar memories onto the image, which could generate diverse imaginations of the external world.

Continuity

Most people's cognition about the city they live in may be composed of a lot of single locations of the city such as the working place, school, market, and home. Besides the static impressions, I want to create dynamic and continuous impressions of living to fill the vacancy between the spots.

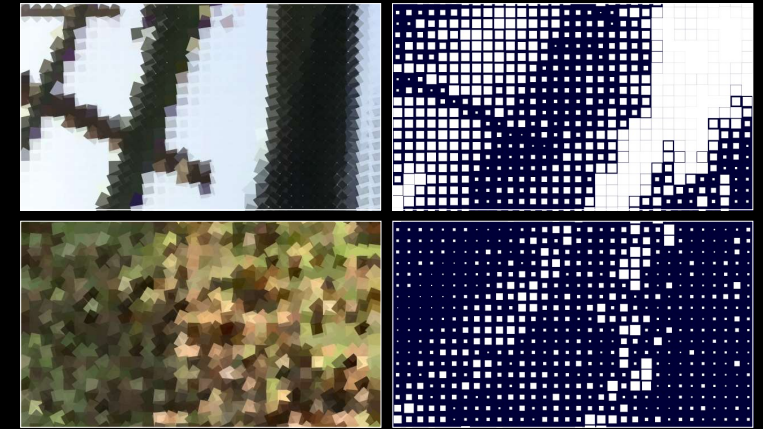


Fig. Frame Pixelation

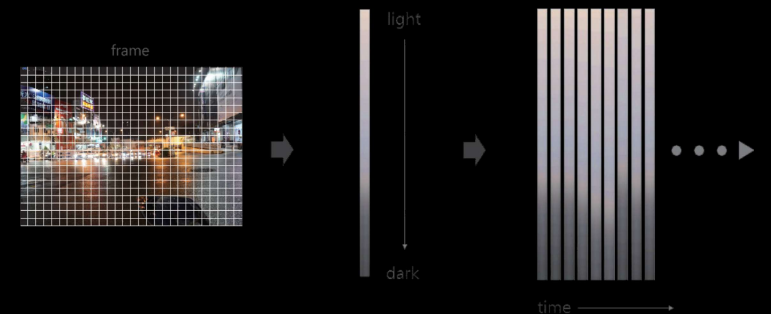


Fig. The Rule of the Translator

2. The Translator

Data Collection

I collected information on the street using the camera that imitates human eyes, then I recorded all the data as an input of my program to compute the results. The process compresses the experience of walking along the Haian Road and translates the images into a new image that contains the time and rich visual impressions.

Translation from videos to images

I stored the RGB color value of every pixels from each frame. I used one frame as a unit, sorted by the brightness from light to dark, then map the value into a rectangle. I piled up each rectangle in sequences. The y-axis presents the brightness of the space and the x-axis presents the time.

By the translation, I tried to deconstruct the general cognition about the city and make the viewers concentrate on the impressive colors of the city by compressing the dynamic video into a two-dimension image.

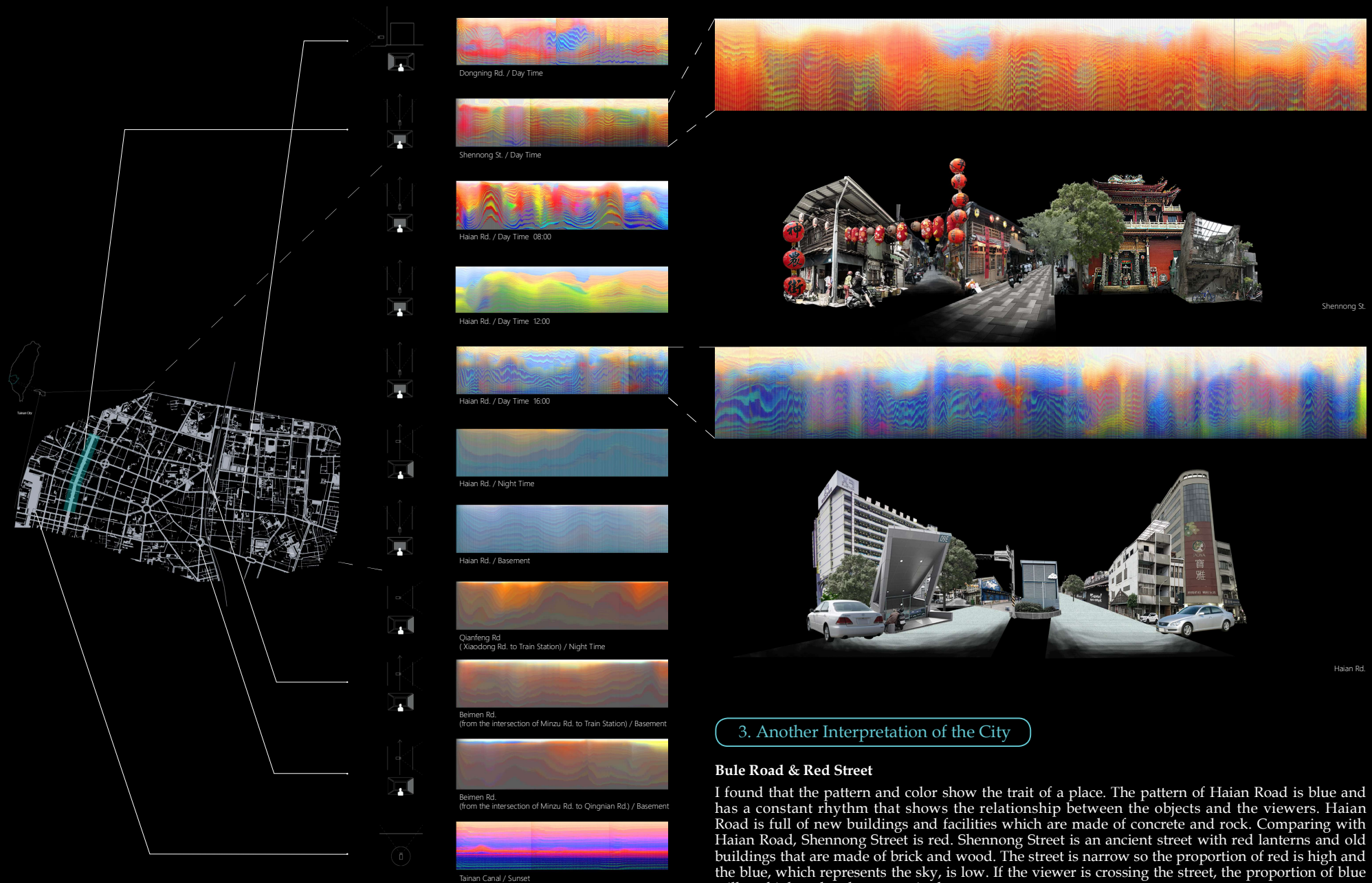


Fig. The Interpretation of Different Street in Tainan City

3. Another Interpretation of the City

Bule Road & Red Street

I found that the pattern and color show the trait of a place. The pattern of Haian Road is blue and has a constant rhythm that shows the relationship between the objects and the viewers. Haian Road is full of new buildings and facilities which are made of concrete and rock. Comparing with Haian Road, Shennong Street is red. Shennong Street is an ancient street with red lanterns and old buildings that are made of brick and wood. The street is narrow so the proportion of red is high and the blue, which represents the sky, is low. If the viewer is crossing the street, the proportion of blue will get high and make a wave in the pattern.

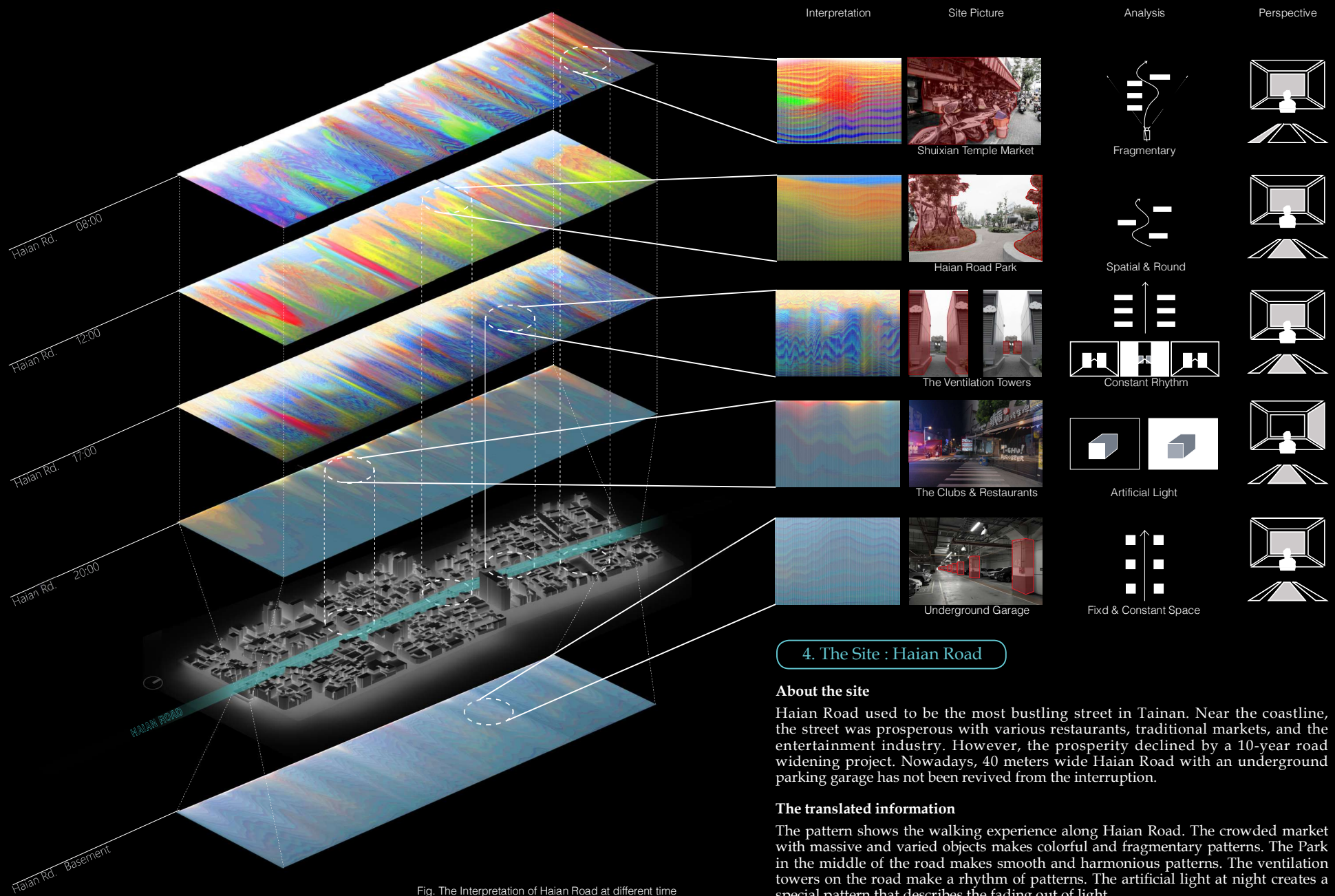


Fig. The Interpretation of Haian Road at different time

4. The Site : Haian Road

About the site

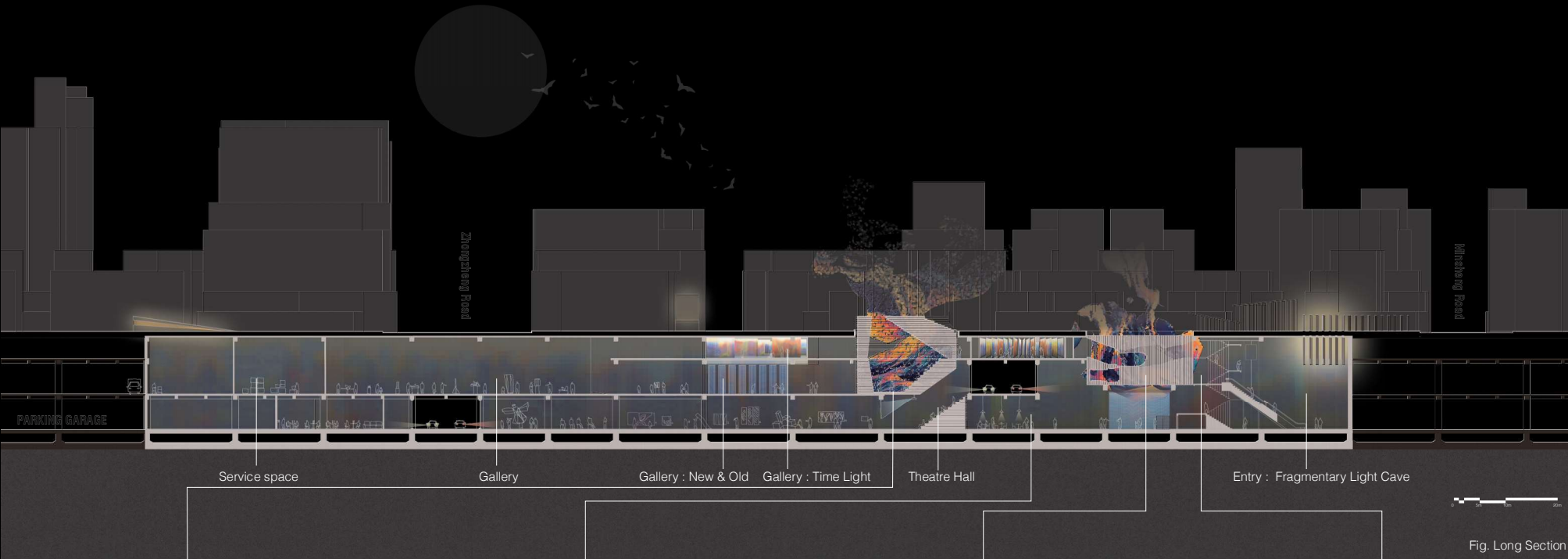
Haian Road used to be the most bustling street in Tainan. Near the coastline, the street was prosperous with various restaurants, traditional markets, and the entertainment industry. However, the prosperity declined by a 10-year road widening project. Nowadays, 40 meters wide Haian Road with an underground parking garage has not been revived from the interruption.

The translated information

The pattern shows the walking experience along Haian Road. The crowded market with massive and varied objects makes colorful and fragmentary patterns. The Park in the middle of the road makes smooth and harmonious patterns. The ventilation towers on the road make a rhythm of patterns. The artificial light at night creates a special pattern that describes the fading out of light.

5. The Underground Observatory

Haian Road contains dual characteristics: bright and spacious above the ground, dark and forgotten under the road. As a compiler, space was put under the Haian Road, in the gap between the bright surface and the dark underground. Experience the leaking light above in the deepest darkness, and redefine the road and the ancient city.



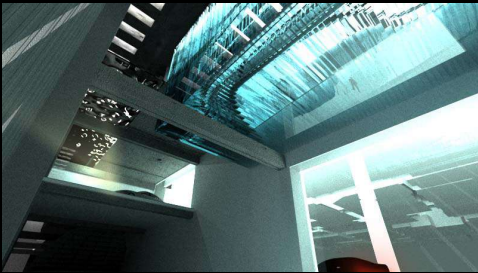
Sealight Plaza

Looking up at the sliced light and city life.



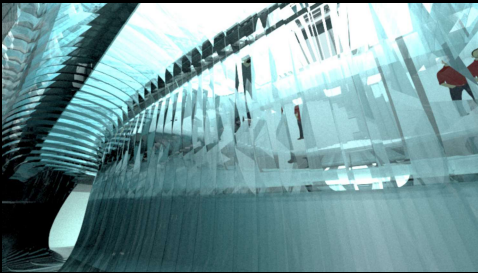
Hall : Flow of Light

The Overlap of light flow



Sliced Light Tunnel

Walking through the compiler to see another view



The Entry of Sliced Light Tunnel

Defferent rhythm of the light

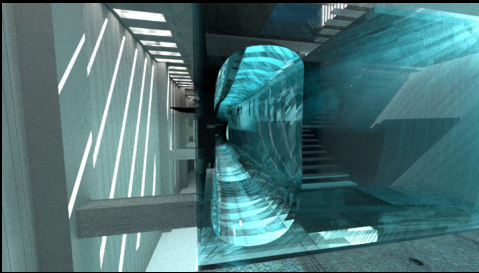


Fig. Narrative Scenarios

03

Melting into River

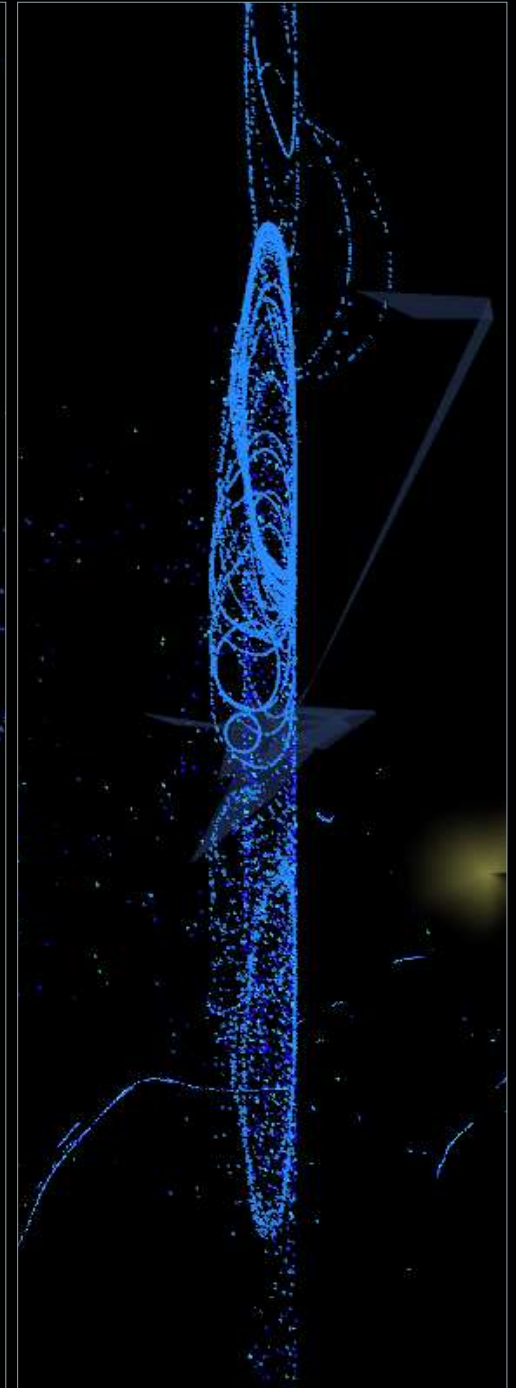
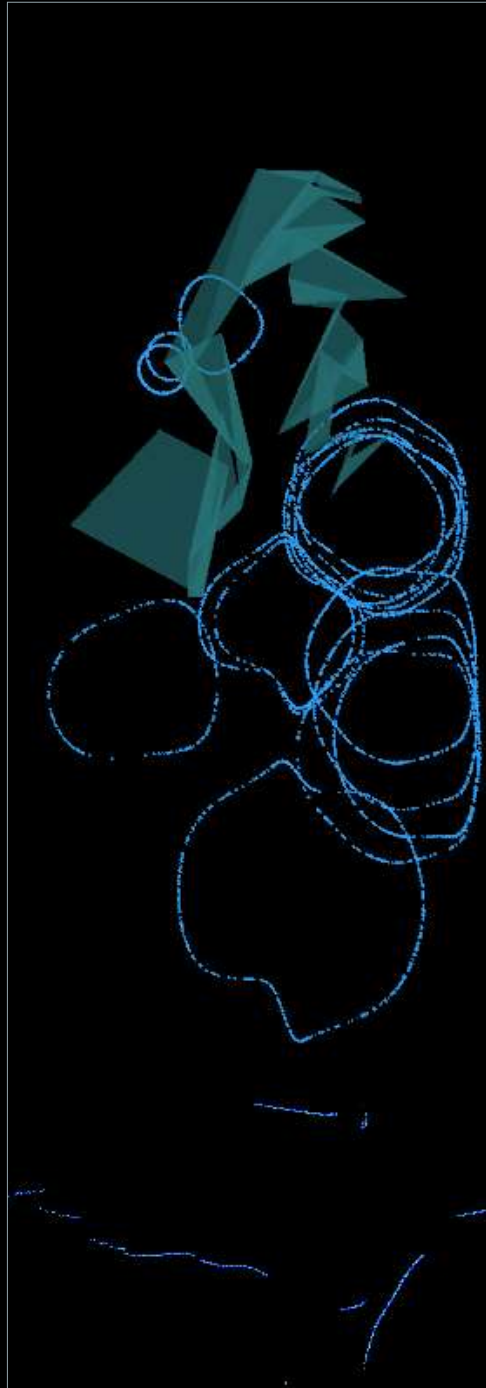
Interactive Design
Expressing the Emotion of Separation

Type | Academic Work, Individual Project

Technique | Java on Processing, Motion Capture by Kinect

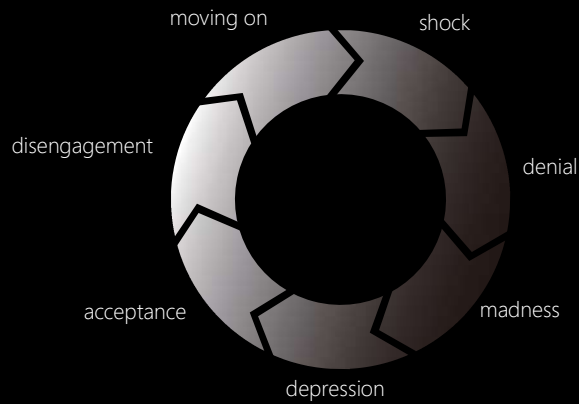
Advisor | Yen-Ting Cho, choyenting@mail.ncku.edu.tw

The project aims to create a complete experience from the interaction, the emotions, the meaning behind the scene, and the special work after it to memorize the experience.

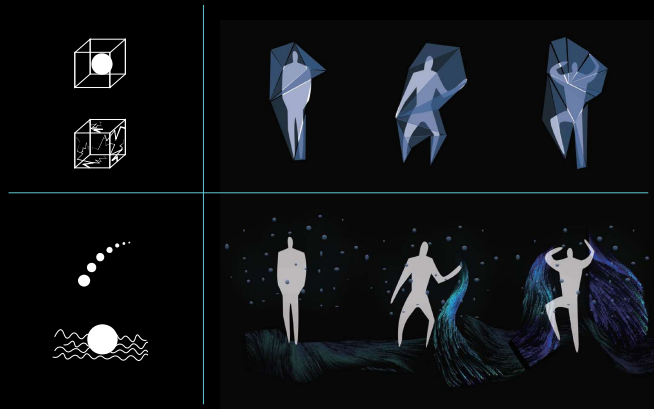


1. Concept

Emotional Cycle



Form concept



Separation has always been a tough issue in people's lives. I studied the emotional changes of people after separating from someone they loved. Some people tend to isolate part of themselves from the outside world like frozen ice. However, they may have a breakdown when being alone at night. The frozen ice finally melts into the water, spreads all the feelings and hurts. Thus, I developed forms inspired by ice and melting and designed several phrases to describe the cycling process of emotional changes.

I divided the design into three sections: the skeleton, the active motion triggered by the participant, and the automatic motion. I developed these three parts separately and organized them together harmoniously.

2. Scenario board



3. Design process

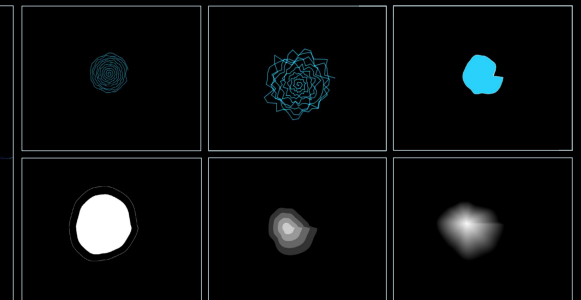
3-1. The skeleton

At first, all the body skeletons touch the automatically dropping snow would trigger the active motion drawing triangles. However, I found that it was ambiguous and rush for the participants to figure out the rule of interaction. The trigger points must be clear and simple to guide the participants on what to do. Thus, I narrowed down the trigger points to only one skeleton point, hand, which are instinct for people to use.

Skeleton points from all to one



Development of skeleton points

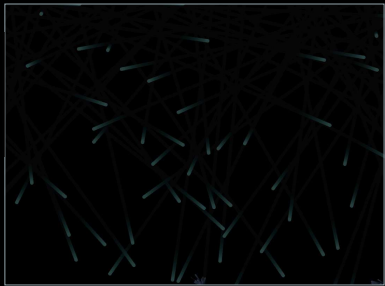


3-2. The automatic motion

Automatic Motion Diagram

Motion	One way	Wind Blow	Dropping	Radial
Form				
Spot				
Linear				
Patch				

Daytime : Dropping Snow



Nighttime : River Flow

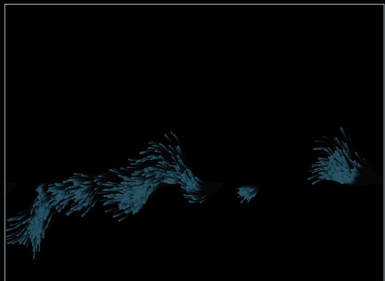
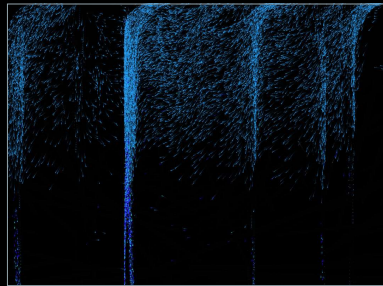
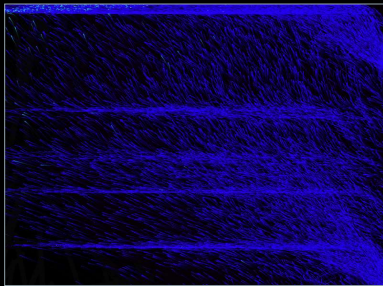
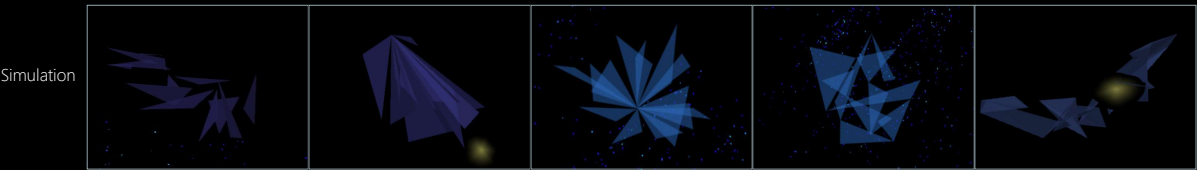


Fig. Simulation on Processing

3-3. The triggered motion

Daytime : Frozen ice

	Intercrossed	Umbrella	Radial	Fragmentary	Folded
Surface					
Frame					



Nighttime : Melting

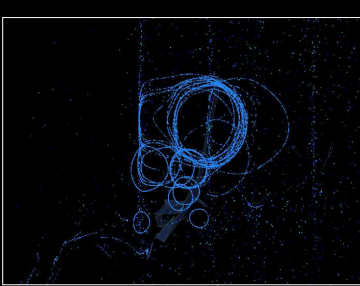
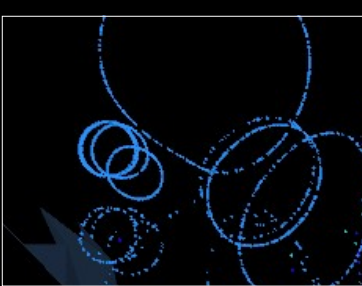


Fig. Simulation on Processing

The Automatic Motion

Inspired by raindrops, I created lots of particles and designed the way how particles move to simulate the rain and the flow of the river. In the scene, people immerse the sadness and feel down. I hope the participant could feel peace through the rain and flow.

The Triggered Motion

I tried different forms combination of triangles to imply the broken heart and frozen mind. At nighttime, the ice melts like a ripple fade away and away to imply sadness away.

4. Live Interaction

Hearing the rain, the participant touched the raindrops and gradually froze himself. In the darkest night, the frozen heart melted into a river.

During the interaction, the participant could be able to immerse in the atmosphere of the story and feel the emotion from it. Moreover, I utilized the movement interacting with the virtual world to create visual patterns able to be printed after the experience.



Fig. Interactive process

04

Bubble Up

A Dome Flagship for the Gas Station

Type | Academic Work

Advisor | Yi-Hsuan Tu, yhtu@mail.ncku.edu.tw

Group Project | Partner_Shao-Fen Chiou

Contribution | Concept, Design and Details, Form Study, Drawing, Presentation.

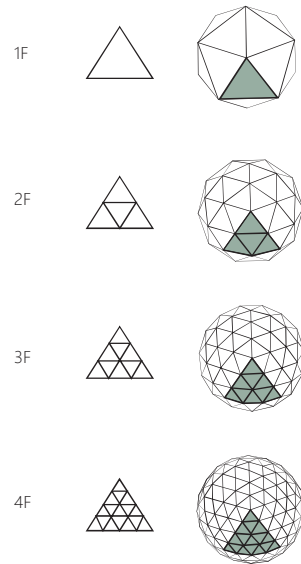


1. Form Study

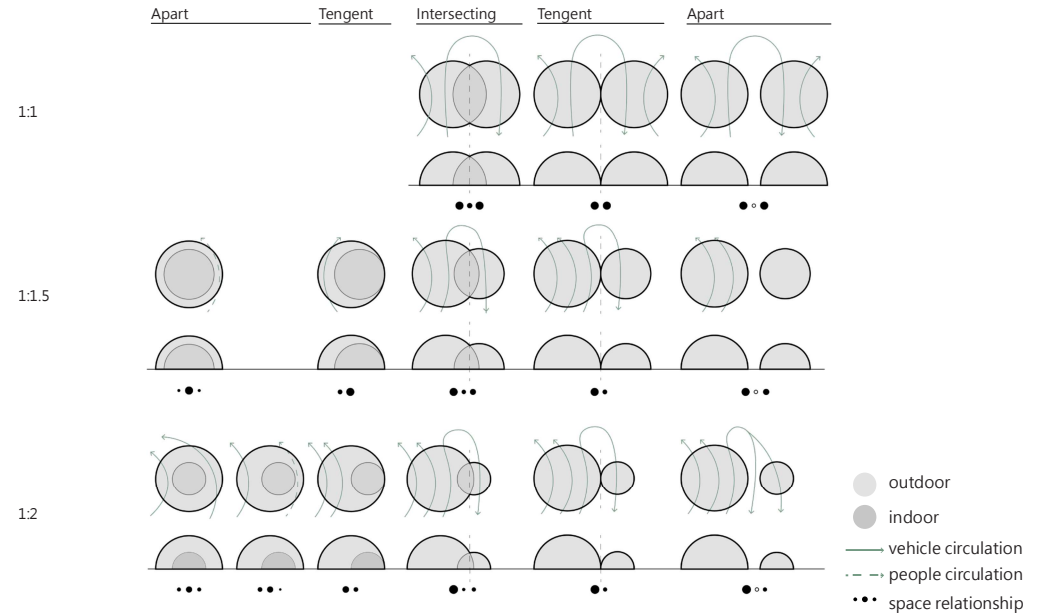
This project aims to design a flagship for a gas station brand, Smile. Besides as a gas station, Smile is famous for its high-quality service of car washing and waxing. Hence, inspired by the bobble of washing, we tried to combine the structure of the building and the figure of Smile.

After studying the geometric relationship of circles, I mainly used two intersecting circles to separate three different functions of space: outdoor space of gasoline, car-washing, waxing, and indoor office.

Structure : Fuller's Dome



Typology and Circulation Relationships of Domes



2. Site

The site is located at the point between the road connected to other cities and the central city. Also, the station is at the end of the road. In sum, the site is a perfect spot for a flagship store.

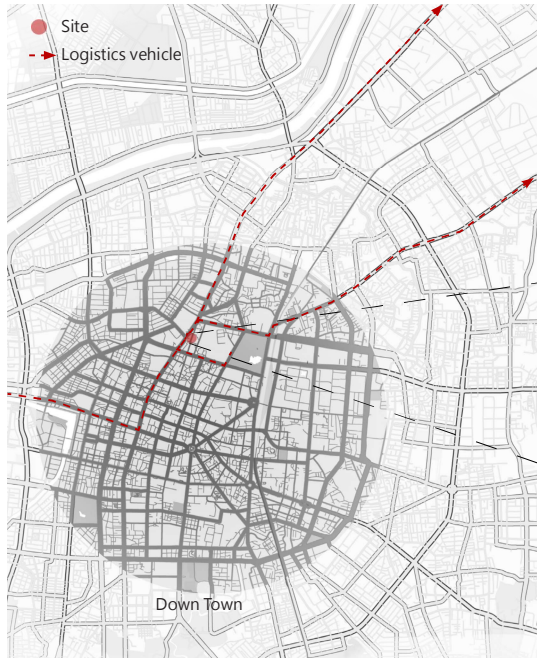


Fig. Tainan City

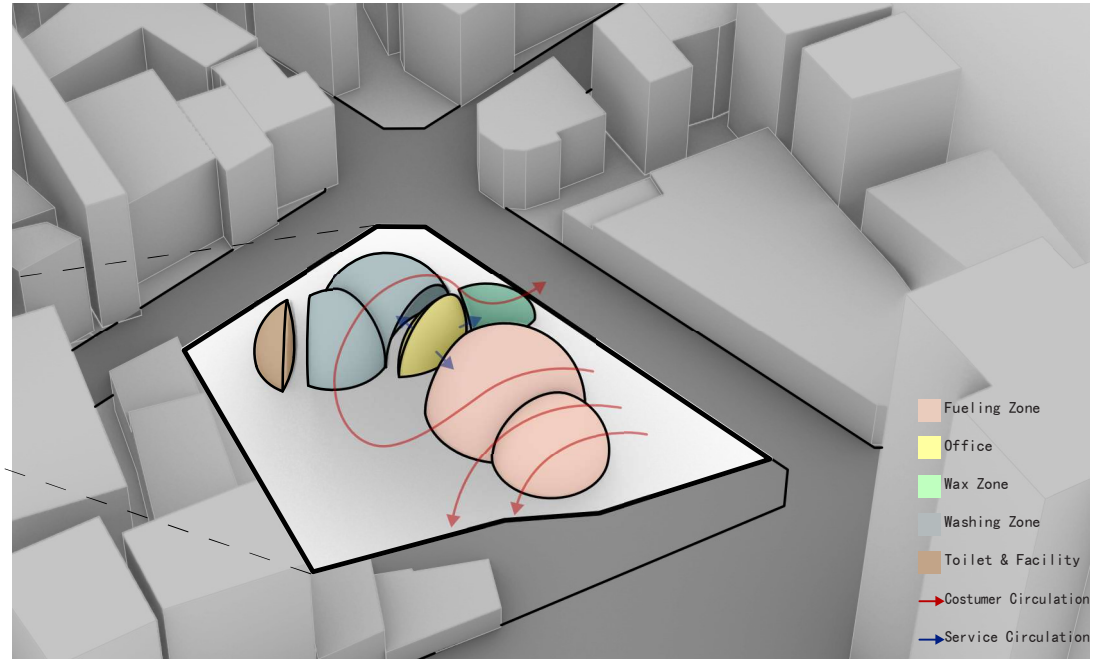


Fig. Program & Circulation 20

3. Space

The office between the two different types of programs, gasoline and car-washing, aims to differentiate them and make a sufficient circulation for service support. West of the site is Ximen Road which is connected to the highway, logistics vehicle and tourist car often pass by. Hence, besides the general cars, there is an extended heightened shed for oversized vehicles.

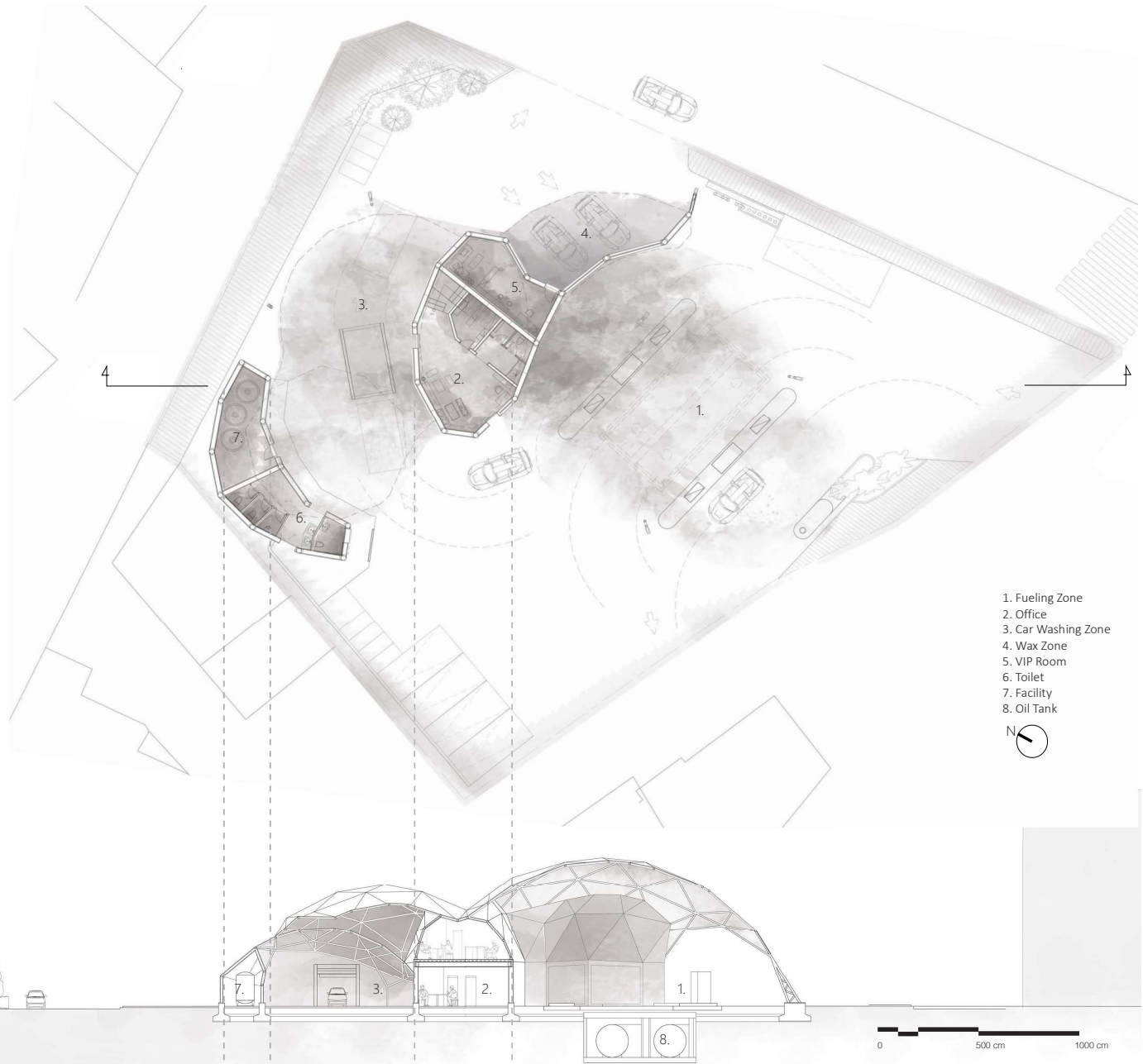
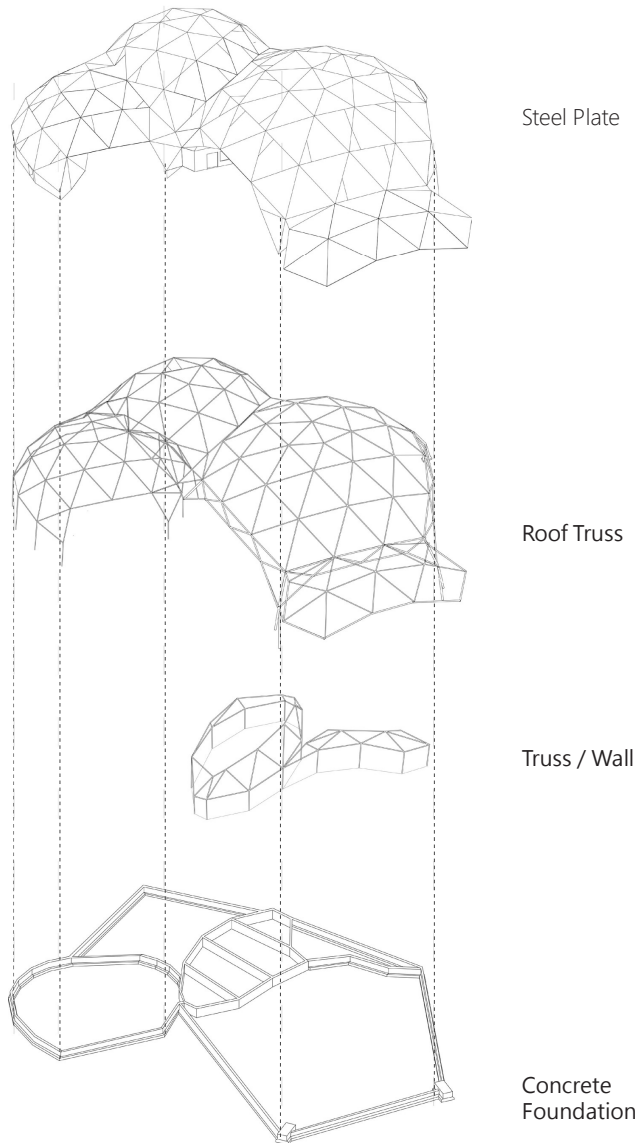


Fig. Plan & Section
The relationship of car-washing zone, office, and waxing zone

Structure explosion model



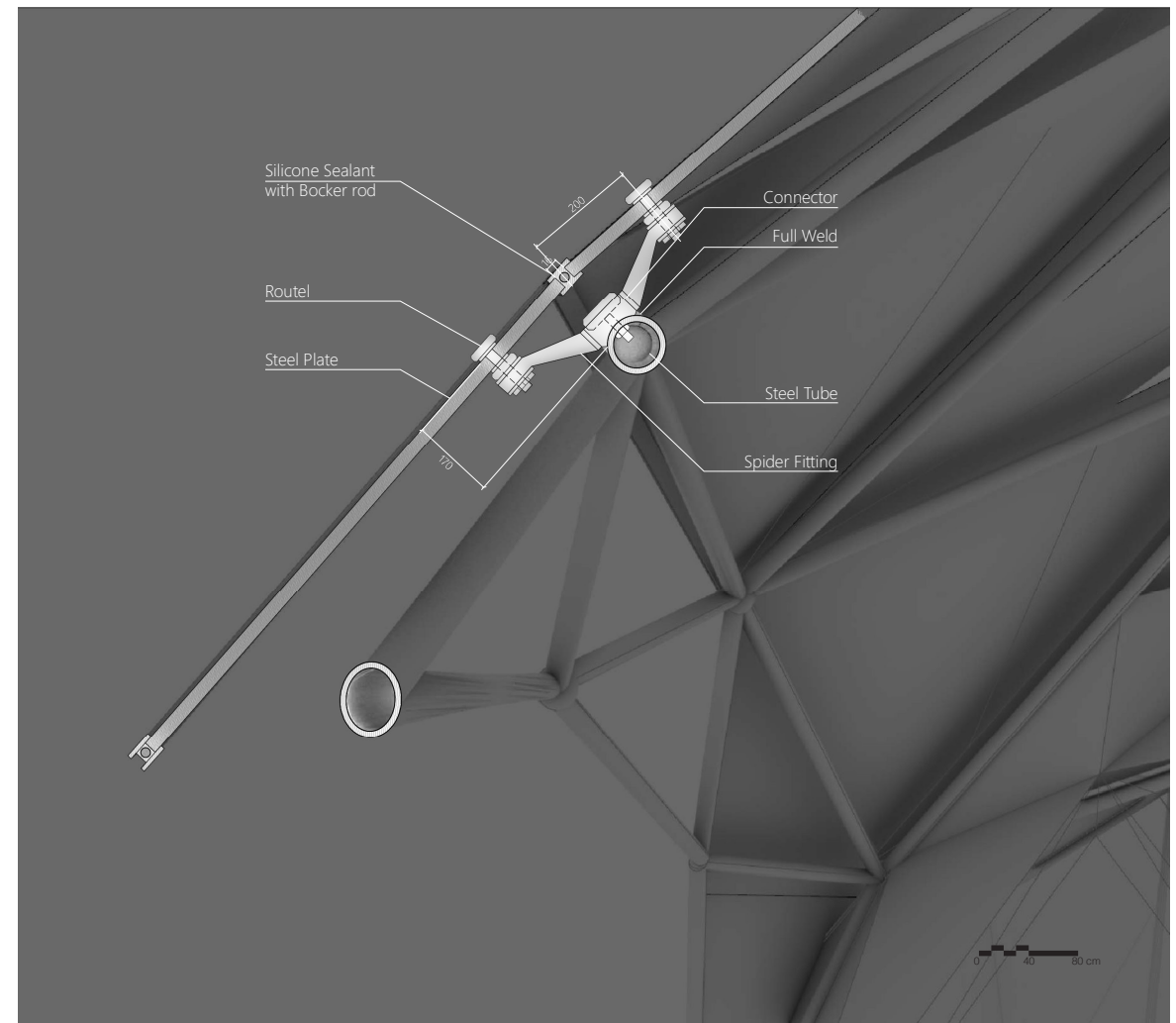
4. Structure

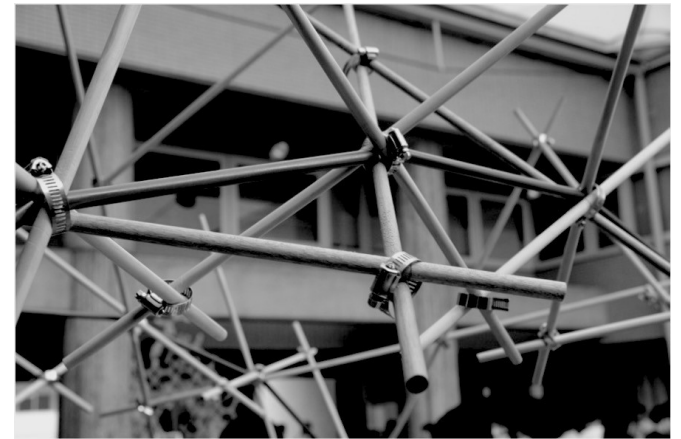
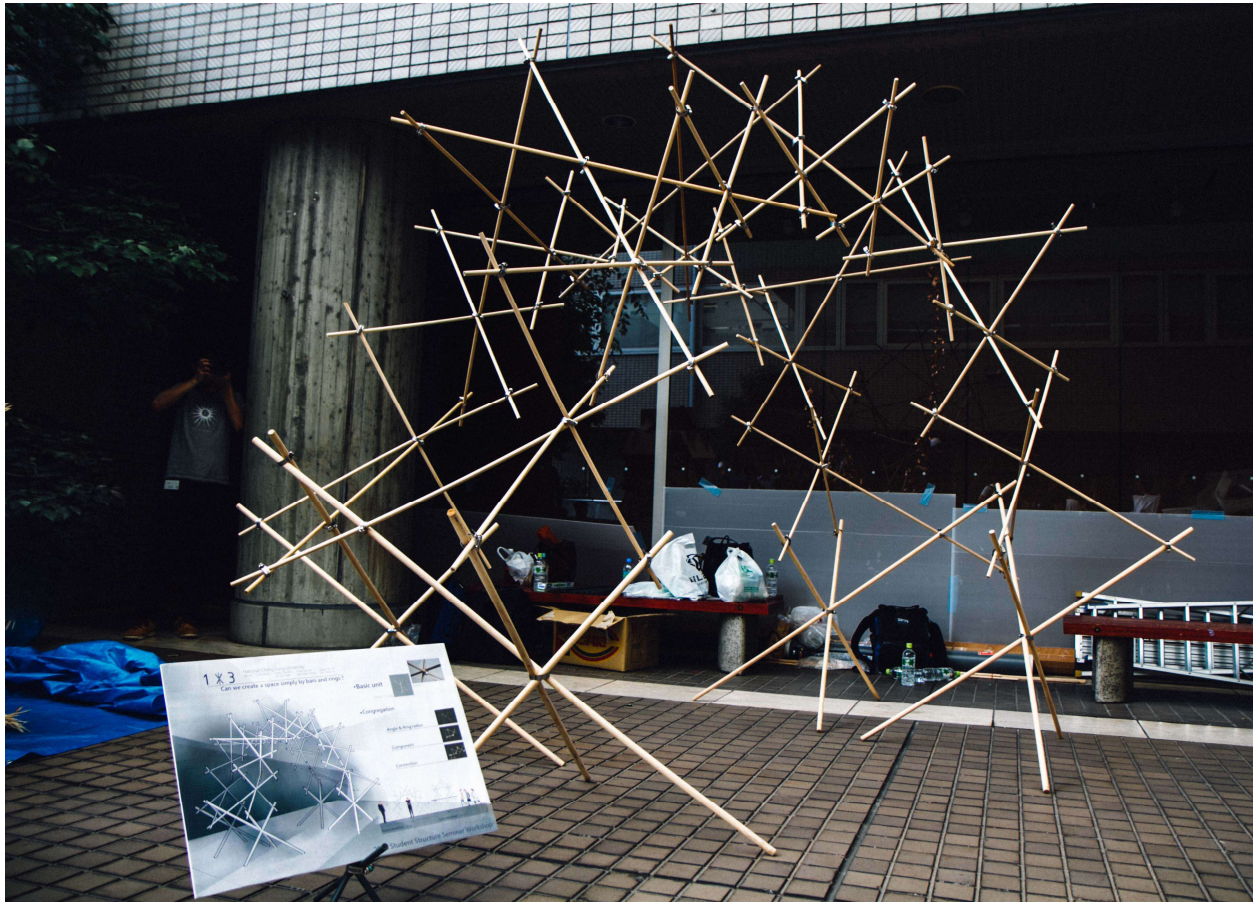
We utilized the rule of the fuller dome to adjust the trusses fitting the demand of different spaces. The intercrossed part is designed as a two stories indoor space, which needs a deeper foundation structure. To reduce the self-weight, the second store and the extended shed must use lighter, small size trusses.



Fig. The perspective of fueling zone

Details





05

1 x 3

2016 Student Summer Seminar Structural Design Competition

Award | 2nd Prize

Advisor | Yi-Hsuan Tu, yhtu@mail.ncku.edu.tw

Group Project | Group of 6

Contribution | Concept, Structure Development, Construction

We designed a new form combining structure, material, and aesthetics. After accumulation, a simple unit turned into a complete structure with various changes.

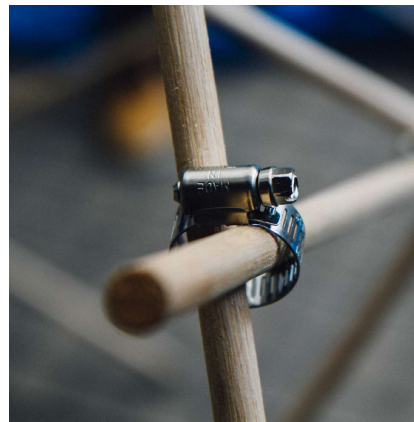
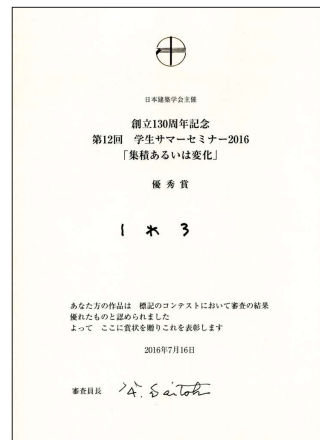


Fig. The connection

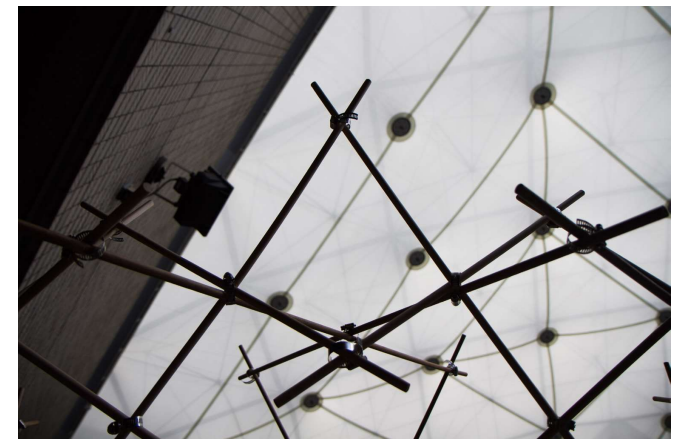
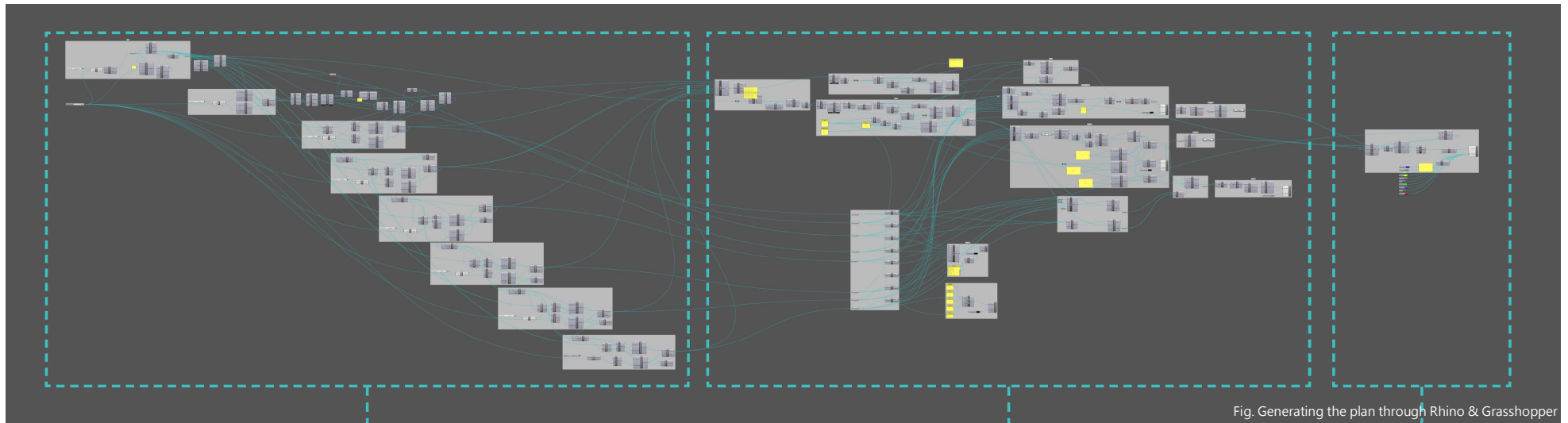


Fig. The construction process 23



Organizing the space in sequence.

Drawing the architectural plan

Drawing the color blocks

We analyzed the spatial relationship of the twelve samples of standard residential houses of the government. To expand the database, we used grasshopper to generate plans following the rules: the plan begins from the gate, hallway, stairs to different programs of spaces and we arrange the space in a specific sequence.

06

Explorations of Plans Generator by Machine Learning

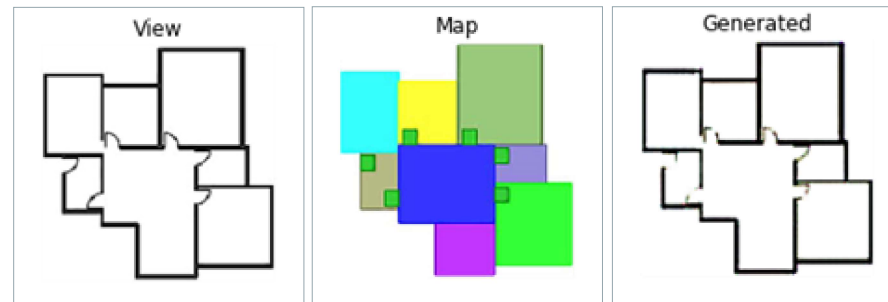
Group Members | Hao Hsu, Cheng-Lin Chuang, Chia-Fan Hsu

Technique | Machine Learning / Pix2pix / GAN

Contribution | The Logic of spatial organization, Generating Plans with Grasshopper, Training Model

This project aims to generate architectural plans fitting specific construction rules by machine learning to assist the architectural design process.

In future use, the model is suitable to utilize at residential houses, factories, or any building with clear and specific construction laws. The designer could simply give the basic data to generate several plans fitting the law to assist the designer in accelerating the design process. Moreover, it may able to summarize the style of a studio which could assist designers to develop the design style and communicate with others more efficiently.



The Model

We used the pix2pix method to tell different spaces and train the model to generate a plan through generative adversarial network.

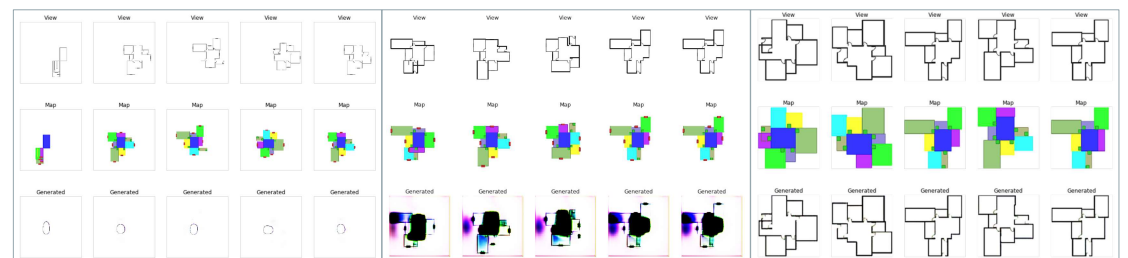
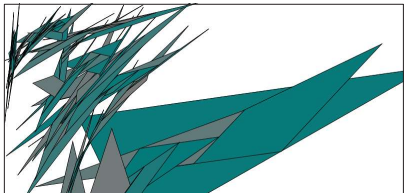
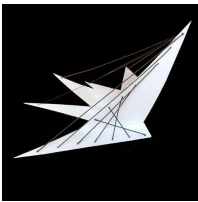
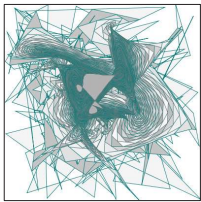
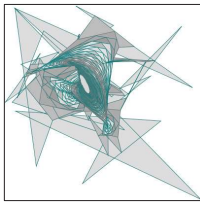
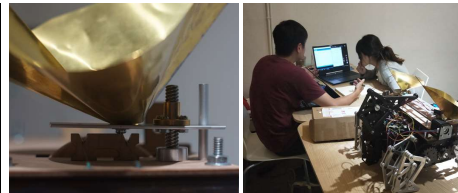


Fig. The training process of the model 24



Explorations of Digital and Manual Computations in Design
Independent Study

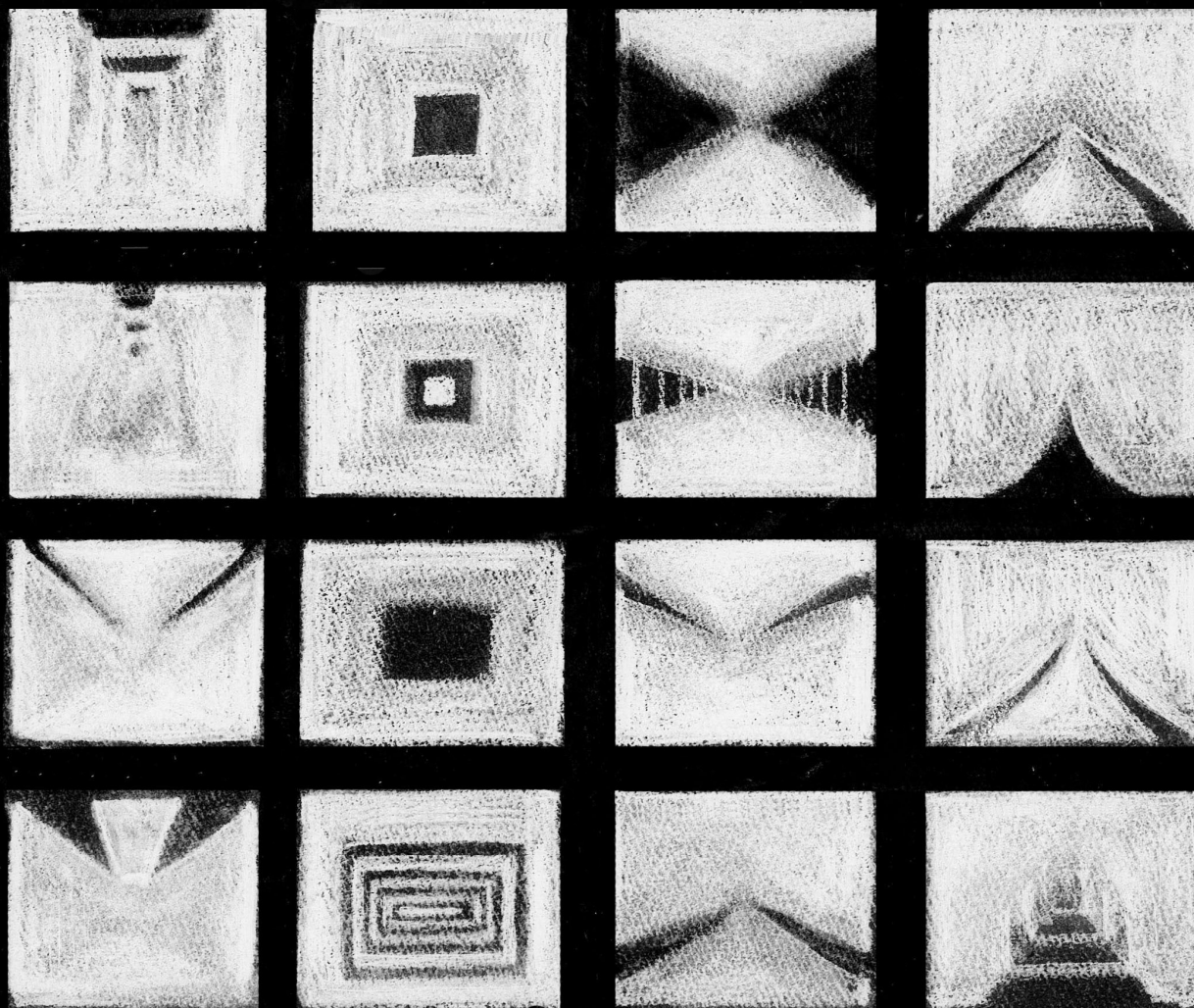
Light Painting
Using Arduino to create a responsive facade



BUGs
Internship Experience, Musical with mechanical bugs

Outro

Beginning with my independent study, I developed a new way to explore the world. In "Light Painting", I learned how to bring designs into reality with aesthetic. In "Bugs", the audience could project themselves onto the cold, solid mechanical bugs and understand the abstract philosophical concept about the struggle of the human body and soul. It was very touching to see how people interact with technology and acquire new ways to navigate and understand the constantly evolving world around them.



YEN-YI HUANG